

EXP-13

Aim

To write a Pandas program to detect the missing values in a given Data Frame and display the result as True or False.

Algorithm

1. Start the program.
2. Import the pandas and numpy libraries.
3. Create a DataFrame containing some missing (NaN) values.
4. Display the original DataFrame.
5. Use the isnull() function to identify missing values.
6. Display the Boolean DataFrame where:
 - **True** indicates a missing value.
 - **False** indicates a non-missing value.
7. Stop the program.

Code :

```
import pandas as pd
import numpy as np

# Creating the DataFrame
data = {
    'ord_no': [70001, np.nan, 70002, 70004, np.nan, 70005, np.nan, 70010, 70003, 70012, np.nan, 70013],
    'purch_amt': [150.50, 270.65, 65.26, 110.50, 948.50, 2400.60, 5760.00, 1983.43, 2480.40, 250.45, 75.29, 3045.60],
    'ord_date': ['2012-10-05', '2012-09-10', np.nan, '2012-08-17', '2012-09-10', '2012-07-27',
                 '2012-09-10', '2012-10-10', '2012-10-10', '2012-06-27', '2012-08-17', '2012-04-25'],
    'customer_id': [3002, 3001, 3001, 3003, 3002, 3001, 3001, 3004, 3003, 3002, 3001, 3001],
    'salesman_id': [5002, 5003, 5001, np.nan, 5002, 5001, 5001, np.nan, 5003, 5002, 5003, np.nan]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nMissing Values (True = Missing, False = Not Missing):")
print(df.isnull())
```

Output :

Original DataFrame:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001.0	150.50	2012-10-05	3002	5002.0
1	NaN	270.65	2012-09-10	3001	5003.0
2	70002.0	65.26	NaN	3001	5001.0
3	70004.0	110.50	2012-08-17	3003	NaN
4	NaN	948.50	2012-09-10	3002	5002.0
5	70005.0	2400.60	2012-07-27	3001	5001.0
6	NaN	5760.00	2012-09-10	3001	5001.0
7	70010.0	1983.43	2012-10-10	3004	NaN
8	70003.0	2480.40	2012-10-10	3003	5003.0
9	70012.0	250.45	2012-06-27	3002	5002.0
10	NaN	75.29	2012-08-17	3001	5003.0
11	70013.0	3045.60	2012-04-25	3001	NaN

Missing Values (True = Missing, False = Not Missing):

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	False	False	False	False	False
1	True	False	False	False	False
2	False	False	True	False	False
3	False	False	False	False	True
4	True	False	False	False	False
5	False	False	False	False	False
6	True	False	False	False	False
7	False	False	False	False	True
8	False	False	False	False	False
9	False	False	False	False	False
10	True	False	False	False	False
11	False	False	False	False	True

Result

The missing values in the given DataFrame were successfully detected using the **isnull()** function.

The output displayed **True** for missing (NaN) values and **False** for all non-missing values.

EXP-14

Aim

To write a Pandas program to find and replace the missing values in a given DataFrame that do not have any valuable information.

Algorithm

1. Start the program.
2. Import the pandas and numpy libraries.
3. Create a DataFrame containing missing values (NaN, "?", and "--").
4. Display the original DataFrame.
5. Replace "?" and "--" with NaN using the replace() function.
6. Fill the missing values using the fillna() function.
7. Display the updated DataFrame.
8. Stop the program.

Code:

```
import pandas as pd
import numpy as np

# Create DataFrame
data = {
    'ord_no': [70001, np.nan, 70002, 70004, np.nan, 70005, "--", 70010, 70003, 70012, np.nan, 70013],
    'purch_amt': [150.5, 270.65, 65.26, 110.5, 948.5, 2400.6, 5760, "?", 12.43, 2480.4, 250.45, 3045.6],
    'ord_date': ["?", "2012-09-10", np.nan, "2012-08-17", "2012-09-10",
                 "2012-07-27", "2012-09-10", "2012-10-10",
                 "2012-10-10", "2012-06-27", "2012-08-17", "2012-04-25"],
    'customer_id': [3002, 3001, 3001, 3003, 3002, 3001,
                    3001, 3004, "--", 3002, 3001, 3001],
    'salesman_id': [5002, 5003, "?", 5001, np.nan, 5002,
                    5001, "?", 5003, 5002, 5003, "--"]
}

df = pd.DataFrame(data)

print("Original DataFrame:\n")
print(df)

# Replace '?' and '--' with NaN
df.replace(["?", "--"], np.nan, inplace=True)

# Fill missing values
df["ord_no"] = df["ord_no"].fillna(0)
df["purch_amt"] = pd.to_numeric(df["purch_amt"]).fillna(0)
df["customer_id"] = pd.to_numeric(df["customer_id"]).fillna(0)
df["salesman_id"] = pd.to_numeric(df["salesman_id"]).fillna(0)
df["ord_date"] = df["ord_date"].fillna("Not Available")

print("\nDataFrame after replacing missing values:\n")
print(df)
```

Output :

Original DataFrame:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.5	?	3002	5002
1	NaN	270.65	2012-09-10	3001	5003
2	70002	65.26	NaN	3001	?
3	70004	110.5	2012-08-17	3003	5001
4	NaN	948.5	2012-09-10	3002	NaN
5	70005	2400.6	2012-07-27	3001	5002
6	--	5760	2012-09-10	3001	5001
7	70010	?	2012-10-10	3004	?
8	70003	12.43	2012-10-10	--	5003
9	70012	2480.4	2012-06-27	3002	5002
10	NaN	250.45	2012-08-17	3001	5003
11	70013	3045.6	2012-04-25	3001	--

DataFrame after replacing missing values:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	Not Available	3002.0	5002.0
1	0	270.65	2012-09-10	3001.0	5003.0
2	70002	65.26	Not Available	3001.0	0.0
3	70004	110.50	2012-08-17	3003.0	5001.0
4	0	948.50	2012-09-10	3002.0	0.0
5	70005	2400.60	2012-07-27	3001.0	5002.0
6	0	5760.00	2012-09-10	3001.0	5001.0
7	70010	0.00	2012-10-10	3004.0	0.0
8	70003	12.43	2012-10-10	0.0	5003.0
9	70012	2480.40	2012-06-27	3002.0	5002.0
10	0	250.45	2012-08-17	3001.0	5003.0
11	70013	3045.60	2012-04-25	3001.0	0.0

Result

The missing values (NaN, "?", and "--") in the DataFrame were successfully identified and replaced using the replace() and fillna() functions. The cleaned DataFrame was displayed successfully.